STAT 342

# Say Yes to the TEST

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Welcome to Kleinfeld Bridal! I'm Randy. We have some statistical conundrums we are facing at the boutique. Let's review the content in STAT 342 and solve these issues!






Let’s review
the Z-test!

# Z-test

***Z-tests are used to decide if a sample mean is significantly different than the true population mean***

## Z-test Requirements

- Used to compare $\bar{x}$ to the true population mean.
- Randomly sampled
- If $\sigma$ is known

## Central Limit Theorem

The CLT guarantees that if n is large, $\bar{x}$ is normally distributed with these parameters.

$$\hat{\mu}_0 = \bar{X} \approx Norm\left(\mu_0, \frac{\sigma_0}{\sqrt{n}}\right)$$

## Confidence Interval

A 100(1-$\alpha$)% confidence interval estiamtor for $\bar{x}$ is:

$$\bar{X} \pm z_{\alpha/2}\frac{\sigma_0}{\sqrt{n}}$$

## Test Statistic

The test is based on the sample mean, $\bar{x}$.

$$Z = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0} \approx Norm(0, 1)$$



Here’s the issue! I’m sure that the average price of a wedding dress at Kleinfeld is $3,000. However, a bride is arguing that she thinks it’s higher than $3,000. Let’s check out who’s right in the next slide.

# Example

Kleinfeld claims that their average price of a wedding dress they sell is **$3,000.**
With nationwide data, we know that the population standard deviation of wedding dress prices is $\sigma$ **= $300.**

A shopper suspects that Kleinfeld is more expensive than the other boutiques she's gone to.

She randomly tries on **24 dresses** and the average price from these dresses is **$3,425.**

## Question

Are the dresses from Kleinfeld more expensive?

## Hypotheses

H0: μ= 3,000

Ha: μ> 3,000

## Test Statistic

$$Z = \frac{\sqrt{24}(3425-3000)}{300} = 6.940$$

## P-value

P(Z ⩾ 6.940) = 1 - pnorm(6.940) =

**1.96×10^{-12}**

We reject the null because the p-value is **Less** than $\alpha$ = 0.05.

## Confidence Interval

$$3425 \pm 1.96 * \frac{300}{\sqrt{24}} =$$

$$[3305, 3545]$$

## Conclusion

At a 0.05 significance level, there is statistically significant evidence that the average price of wedding dresses at Kleinfeld is greater than $3,000!
Also, we are 95% confident that the true average price of wedding dresses at Kleinfeld lies between $3,305 to $3,545.

The customer was right! The average cost of a wedding dress at the boutique is from  $3,305 to $3,545!





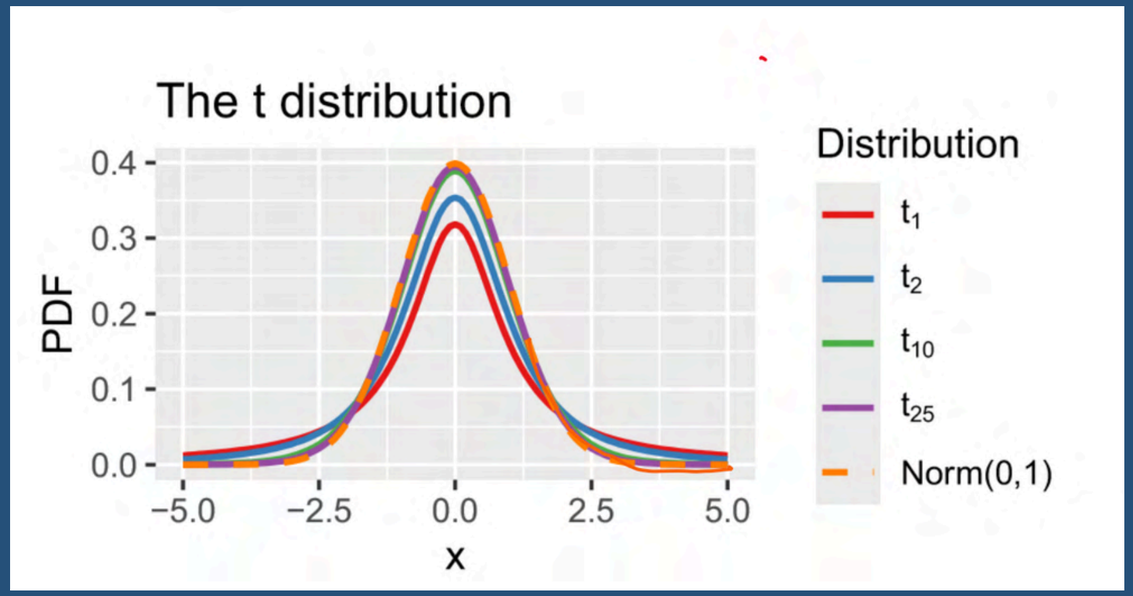
Now let’s dive into the T-distribution and its uses in statistical testing!

# What is the T-Distribution?

***What happens when we are faced with a too-small sample size to assume the Central Limit Theorem holds and use a Z-test? We use the T-Distribution!***

## T-Distribution Properties

- Bell shaped and symmetric about 0
- The PDF depends on the degrees of freedom
- Thicker tails and more spread out than a normal distribution
- Converges to a normal distribution as the degrees of freedom increase




## When is it useful?

We've already established the Z-test, which uses the test statistic:

$$Z = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0} \approx Norm(0, 1).$$

Since population standard deviation is almost always unknown, we generally estimate the true population sd σ0 as the sample sd S. However, for small n values, the ratio above will no longer follow a standard normal distribution, but rather a T-Distribution:

$$T = \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \sim t_{n-1}$$

# Elements of the T-Test for One Sample:

Suppose $X_1, X_2, ..., X_n$ are drawn independently from a $Norm(\mu_0, \sigma_0)$ distribution, with both parameters unknown. Then:

$$T = \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \sim t_{n-1}$$

Where n - 1 denotes the degrees of freedom of the distribution. We can use this test statistic and its distribution for making inference about the true population mean μ0.

For this case, a 100(1 - α)% confidence interval for μ0 is the set of values:

$$\bar{X} \pm t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}$$

The test which uses the T ratio to conduct a significance test for the population mean is called a one sample T-Test.

# One Sample T-Test Example

Randy insists that the bride from the earlier Z-test example testing the average price of Kleinfeld wedding dresses redo her statistical test using a one sample T-Test!

Randy thinks it is not accurate to use nationwide data for the standard deviation of wedding dress prices, and demands her to calculate a sample sd for her test. He also believes that her sample size of n = 24 is too small for the Z-test using a sample sd, and therefore a T-Test is the better option.

The shopper had tried on 24 dresses randomly and found an average cost of $3,425. She also finds a sample standard deviation of $292.

## Question

Are the dresses from Kleinfeld more expensive than Randy claims?

## Hypotheses

H0: μ= 3,000

Ha: μ> 3,000

## Test Statistic

$$T = \frac{\sqrt{24}(3425 - 3000)}{292} = 7.13036$$

## P-value

Assuming the null hypothesis is true, our test statistic would be a draw from a T distribution with df = 23

P(T >= 7.13036) = pt(7.13036, 23, lower.tail = F) = 1.46 * 10^{-7}

## Confidence Interval

$$3425 \pm 2.0687 \cdot \frac{292}{\sqrt{24}}$$

$$[3301.7, 3548.3]$$

## Conclusion

Once again we conclude that at a 0.05 significance level, there is statistically significant evidence that the average price of wedding dresses at Kleinfeld is greater than $3,000!

Also, we are 95% confident that the true average price of wedding dresses at Kleinfeld lies between $3,302 to $3,548.



It looks like I was wrong.... again! We have evidence to believe that the average price of a wedding dress at kleinfeld is greater than $3000.

Now lets turn to the Paired T-Test. Sometimes, it’s not about comparing different dresses, it’s about seeing how the same dress changes.



# Elements of the T-Test for Paired Data:

Studying pre intervention and post intervention on the same subject is an example of a paired observation. Suppose:

$$(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n).$$

Are paired observations, and define the differences:

$$D_i = X_i - Y_i, \quad i = 1, \ldots, n$$

$$D_1, D_2, \ldots, D_n \overset{i.i.d.}{\sim} Norm(\mu_d, \sigma_d)$$

with both parameters unknown

μd = μX - μy

σd = $\sqrt{}$ Var(D)

Test Statistic:

$$T_0 = \frac{\sqrt{n}(\bar{D} - \mu_d^{null})}{S} \sim t_{n-1}$$

Where:

D = sample mean of the differences

S = sample SD of the differences

With n-1 degrees of freedom.

Think of a paired t-test as a one-sample t-test on the differences!

# Paired T-Test Example

Randy explains that alterations can significantly change how a dress is valued. He recommends the bride to compare the same 24 dresses before and after alterations using a paired T-Test.

Since the same dresses are being evaluated twice, treating them as independent samples would be inaccurate. Because the population SD of the price differences is unknown and the sample size is small, a paired T-Test is the best option.

The bride records prices before and after alterations with 24 dresses. She observes a mean difference of $180 and a sample SD of differences of $292

## Question

do alterations increase the avereage price of wedding dresses at Kleinfeld?

## Hypotheses

HO: μd= 0

Ha: μd> 0

## Test Statistic

$$T = \frac{\sqrt{24} \cdot 180}{292} \approx 3.02$$

## P-value

Assuming the null hypothesis is true, our test statistic would be a draw from a T distribution with df = 23

$$P(T \geq 3.02) = \text{pt}(3.02, 23, \text{lower.tail} = \text{FALSE}) \approx 0.003$$

## Conclusion

With a signifiance level of 0.05, we can reject the null hypothesis that there is statistically significant evidence that alterations increase the average price of wedding dresses at Kleinfeld.

The bride's observed mean difference of $180 suggests that, on average, alterations add value to the dresses.

Hmm, alterations really do increase the average price of these dresses. Who knew a little tailoring could make such a difference?




Let’s review
the Chi-test!

# Chi-Square

***What if population variance is unknown?***

*Chi-square distribution is from taking the sum of squares of i.i.d normal random variables.*

## Main Properties

The dist is a **continuous** probability ranging from [0, ∞]

The shape of the dist depends on the degrees of freedom (χ2.)

More degrees = PDF is closer to **normal** distribution.

**E[X] = d**
**Var[X] = 2d**

The sum of independent chi square random vars are chi square distribution.

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

**$S^2$** is the sample variance/**unbiased estimator** used in place of unknown $\sigma^2$.

## Confidence Interval

**100(1-α)% confidence interval estimator for $\sigma^2$**

$$\left[\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right]$$

## Test Statistic

$$Q_0 = \frac{(n-1)\,S^2}{\sigma^2_{0,null}}$$

Here’s the next issue!
Kleinfeld wants to understand the variance of Hayley Paige wedding dresses when the true population variance is unknown! I think that that it is $2,000 but no one believes me because of the first two problems!



# Chi-Square Example

Kleinfeld wants to understand how spread out the prices are for Hayley Paige wedding dresses. Kleinfeld thinks that the designer's wedding dresses **variance** is $2,000.

Let's check if the variance is smaller or larger.
The **sample variance** is $2,800 calculated with **36 dresses** and prices are assumed to be **normally distributed.**

```
q <- 49

lower <- pchisq(49, 35)

upper <- pchisq(49, 35, lower.tail = F)

cat("The lower tail p-value is ",lower,"\n")
cat("The upper tail p-value is ", upper, "\n")

p_val <- 2 * upper
cat("The p-value is ", p_val)

```

```
The lower tail p-value is  0.9416245
The upper tail p-value is  0.05837554
The p-value is  0.1167511
```

## Hypotheses

HO: $\sigma^2 = 2{,}000$

Ha: $\sigma^2 \neq 2{,}000$

## Test Statistic

$$Q_0 = \frac{(35)*2800}{2000} = 49 \sim X^2_{35}$$

## Conclusion

Since the p-value is larger than 0.05, we fail to reject the null and say that there is no significant evidence to prove that the variance is **not** $2,000.


Yes, I was right! They couldn't find enough significant evidence to prove that the true population variance is not $2,000.



What if we have two sample means we are comparing? This calls for a two sample test!

# Two Sample Tests

***When we want to compare two population means μ1 and μ2, we will use μ1 - μ2 as our parameter, and estimate it by the observed difference of the sample means. We have two options for two sample tests!***

## Using the Central Limit Theorem

- For large samples of sizes n and m
- Uses Xbar - Ybar as an estimator with assumed distribution:

$$X_1, X_2, \dots, X_n \sim \mu_1, \sigma_1$$

$$Y_1, Y_2, \dots, Y_m \sim \mu_2, \sigma_2$$

$$\bar{X} - \bar{Y} \approx Norm\left(\mu_1 - \mu_2, \sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}\right)$$

- We use a test statistic of:

$$Z = \frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}} \sim Norm(0, 1)$$

## Using a Two Sample T-Test

- Can be used for small sample sizes n and m
- Assumes that both samples have the same population standard deviation

$$X_1, X_2, \dots, X_n \overset{i.i.d.}{\sim} Norm(\mu_1, \sigma_0)$$

$$Y_1, Y_2, \dots, Y_m \overset{i.i.d.}{\sim} Norm(\mu_2, \sigma_0)$$

- When the population standard deviation is known the distribution of Xbar - Ybar is:

$$\bar{X} - \bar{Y} \approx Norm\left(\mu_1 - \mu_2, \sqrt{\frac{\sigma_0^2}{n} + \frac{\sigma_0^2}{m}}\right)$$

- And the test statistic can be calculated as:

$$\frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sigma_0 \sqrt{\frac{1}{n} + \frac{1}{m}}} \sim Norm(0, 1)$$

# Two Sample T-Test Continued

***What do we do for a two sample T-Test where the population standard deviation is unknown?***

**We discussed how to perform a two sample T-Test with a known population standard deviation, but what about when that value is unknown?**

- For this case, we will use the same test statistic as before, but substituting a sample standard deviation for the true value. Like we saw before when covering a one sample test using the t distribution, this changes the distribution of our test statistic :

$$s_p^2 = \frac{1}{n+m-2}\left[(n-1)\,s_x^2 + (m-1)\,s_y^2\right]$$

$$T = \frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \sim t_{n+m-2}$$

- A 100(1 - α)% t-confidence interval for μ1 - μ2 is then:

$$\bar{X} - \bar{Y} \pm t_{\alpha/2,n+m-2}\, S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$$





Thank you for helping me with these statistical conundrums! Our customers are ***significantly*** more happier!

